The floor tells the robot

A dark marker identifies a delivery stop

Today’s robot challenge

Bright and dark readings

A threshold between the readings

A wide marker and a repeated test

GREEN steady: on **ORANGE flashing: charge** **RED flashing: stop + mentor**

Center button lights. A letter on the display is your program’s output.

Calibration and repeated decisions

```
threshold = 40
reading = color_sensor.reflection(port.D)
if reading < threshold:
    await signal("M")
```

Our prediction

Dark is 12 and light is 76. Is 44 a useful threshold?

What if the threshold is below both readings?

Reading fragment from the complete supplied project

Our robot and build jobs

Driver edits and starts after “clear.”

Navigator predicts and resets the stopped robot.

Observer watches, measures and records.

Other team builds. Everyone rotates jobs.

Robot turns: rotate every 4 minutes. Retests: every 2 minutes.

Evidence from our test

We changed _____.

We predicted _____.

We observed _____.

Our next test would _____.

What we learned and noticed

What did you learn about detecting a marker?

What did you observe on light and dark surfaces?

What would you change in the road or next test?

Save your team project. Store your build. Reset the robot and parts.